Crew quarters module Specification

Document number	SHV-HA-SPEC-0003
Revision	A
Status	SUPERSEDED
Classification	INTERNAL
Author	Ilka Warrick
Approved by	Selene Abaroth
Date	2021-11-02

1. Purpose

This specification defines the performance, interface and verification requirements for the crew quarters module (HA-CRQ) of the Habitation Blocks on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-HA-02 (WBS 1.9.2), owned by Sola Quarrel.

The prime supplier for flight hardware is Chandrila Atmospherics. Responsible engineer: Ilka Warrick.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-HA-DWG-0201	Crew quarters module general arrangement drawing
SHV-HA-BOM-0301	Crew quarters module bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-LS-HA	Interface Control Document: Life Support & Atmospherics / Habitation Blocks
SHV-ICD-HA-AG	Interface Control Document: Habitation Blocks / Hydroponics & Food Production

4. Description

The habitation blocks house the resident crew and their families: quarters, mess halls, the medical centre, schools and recreation decks.

The crew quarters module is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
HA-CRQ-R01	The crew quarters module shall provide a berths of 1,800 ea.	1,800 ea	±1%	Test

5.1 General requirements

ID	Requirement	Verification
HA-CRQ-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
HA-CRQ-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
HA-CRQ-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
HA-CRQ-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
HA-CRQ-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the HA test-report series.

8. Revision history

Rev	Date	Description	Author
A	2021-11-02	Berths set to 1,800 ea (superseded).	Ilka Warrick